TROPICAL CYCLONE FORECAST & RESEARCH EVALUATION REPORT

A Deep Learning Framework for Multi-Source Satellite-Based Tropical Cyclone Analysis

Report Identifier:	6d01987b-805a-431a-b9d1-358b0333eb5c	**Generated At:**	2026-10-14 07:53:51 UTC
Cyclone Identifier:	2016341N10095	**Storm Name:**	VARDAH
Basin / Region:	North Indian Ocean (Bay of Bengal)	**Season / Year:**	2016
Peak Sustained Wind:	75.0 knots	**Min Central Pressure:**	974.0 hPa
Highest IMD Stage:	Very Severe Cyclonic Storm (VSCS)	**Data Provenance:**	NOAA IBTrACS / IMD RSMC Best-Track

Spatiotemporal Model Forecast (ConvLSTM)

Lead Time	Forecast Valid Time (UTC)	Latitude (°N)	Longitude (°E)	Wind (kt)	IMD Stage	Cone (km)
+6 h	2016-12-19 06:00:00	9.47°	50.86°	19.7 kt	Depression (D)	±60.0 km
+12 h	2016-12-19 12:00:00	9.99°	50.73°	19.7 kt	Depression (D)	±85.0 km
+18 h	2016-12-19 18:00:00	10.46°	50.62°	19.9 kt	Depression (D)	±110.0 km
+24 h	2016-12-20 00:00:00	10.89°	50.52°	20.2 kt	Deep Depression (DD)	±135.0 km

Synoptic Track Observations (Recent Records)

Timestamp (UTC)	Latitude	Longitude	Wind Speed	Pressure	Observed Stage
2016-12-12 06:00:00	13.1°N	80.5°E	70.0 kt	978.0 hPa	Very Severe Cyclonic Storm (VSCS)
2016-12-13 00:00:00	12.9°N	78.0°E	35.0 kt	998.0 hPa	Cyclonic Storm (CS)
2016-12-14 00:00:00	12.5°N	74.0°E	25.0 kt	1004.0 hPa	Depression (D)
2016-12-15 00:00:00	11.8°N	68.5°E	30.0 kt	1002.0 hPa	Deep Depression (DD)
2016-12-16 00:00:00	11.0°N	63.0°E	30.0 kt	1002.0 hPa	Deep Depression (DD)
2016-12-17 00:00:00	10.2°N	58.0°E	25.0 kt	1004.0 hPa	Depression (D)
2016-12-18 00:00:00	9.5°N	54.0°E	25.0 kt	1004.0 hPa	Depression (D)
2016-12-19 00:00:00	8.9°N	51.0°E	20.0 kt	1008.0 hPa	Depression (D)

Verified Model Benchmark Comparison (Held-Out Test Set)

Model / Baseline Architecture	Mean Track Error (km)	Intensity MAE (knots)	Intensity RMSE (knots)
Persistence Kinematic Baseline	266.98 km	14.00 kt	15.17 kt
Linear Velocity Extrapolation	190.70 km	9.00 kt	13.23 kt
ConvLSTM Spatiotemporal Net (Proposed)	187.39 km	7.81 kt	8.99 kt

Scientific Disclaimer: This report was automatically compiled by an academic deep-learning research prototype. Forecasts and estimated parameters are for scientific verification and educational demonstration purposes only and DO NOT constitute official meteorological warnings or operational emergency guidance. Consult the India Meteorological Department (IMD) and RSMC New Delhi for official cyclone advisories.